

MA 403 Real Analysis

Homework 9

1. Determine which of the following functions are of bounded variation on $[0, 1]$.

$$(a) f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases} \quad (b) f(x) = \begin{cases} \sqrt{x} \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$$

2. A function f , defined on $[a, b]$, is said to satisfy a uniform Lipschitz condition of order $\alpha > 0$ on $[a, b]$ if there exists a constant $M > 0$ such that

$$|f(x) - f(y)| < M|x - y|^\alpha \quad \text{for all } x \text{ and } y \text{ in } [a, b].$$

If f is such a function, show that $\alpha > 1$ implies f is constant on $[a, b]$, whereas $\alpha = 1$ implies f is of bounded variation on $[a, b]$.

3. A function $f : [a, b] \rightarrow \mathbb{R}$ is said to be *absolutely continuous* on $[a, b]$ if for every $\epsilon > 0$ there is a $\delta > 0$ such that

$$\sum_{k=1}^n |f(b_k) - f(a_k)| < \epsilon$$

for every n disjoint open subintervals (a_k, b_k) of $[a, b]$, $n = 1, 2, \dots$ satisfying $\sum_{k=1}^n (b_k - a_k) < \delta$.

Prove that every absolutely continuous function on $[a, b]$ is continuous and of bounded variation on $[a, b]$.

4. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function which is Riemann-integrable, and let c be any real number.

Show that the function cf is Riemann-integrable on $[a, b]$ and $\int_a^b cf(x)dx = c \int_a^b f(x)dx$.

5. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann-integrable functions. Show that the function $f + g$ is

Riemann-integrable on $[a, b]$ and $\int_a^b (f(x) + g(x))dx = \int_a^b f(x)dx + \int_a^b g(x)dx$.

6. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann-integrable functions such that $f \geq g$.

Show that the function $\int_a^b f(x)dx \geq \int_a^b g(x)dx$.

7. Let $f : [a, b] \rightarrow \mathbb{R}$ be a non-negative continuous function satisfying $\int_a^b f(x)dx = 0$. Prove that $f = 0$.